The intelligent Beehive IoT system with the ability to measure bee activity and honey production

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Abstract

This research configures a group of temperature and humidity sensors inside the beehive to monitor the environmental changes within the hive. Furthermore, a servomotor is installed at the ventilation entrance to control the opening of the ventilation gate, thereby preventing the bees from being exposed to unsuitable growth conditions for an extended period. Additionally, the honey production is observed through the feedback from the Force Sensitive Resistor (FSR) to measure changes in honey yield. Moreover, Laser Sensor and Light Intensity Sensor are used to detect the frequency of bee entry and exit from the hive. The sensor data from each beehive is aggregated using LoRa transmission technology and sent to the Receiver. The Receiver then utilizes NB-IoT to transmit the data to a cloud server, allowing beekeepers to monitor real-time honey production and identify any issues where bees are unable to produce honey in a timely manner.

1 Introduction

To address rural depopulation and an ageing workforce, this research takes advantage of the flourishing development of Agriculture 4.0[1][2]. By uploading sensor data from beehives to a cloud server, beekeepers can remotely monitor hive conditions in real time, meeting the demand for labour-saving measures.

Through the execution and implementation of this study, IoT beehives equipped with multiple sensors outperform traditional beehives, enabling digital management to meet the current need for labour-saving practices. In addition, by monitoring the temperature and humidity conditions of each hive to ensure they match the optimal growth environment, it is possible to observe changes in honey production for each honeycomb. This information allows farmers to make informed decisions about whether to move hives based on past experience, ultimately increasing honey yields. In addition, by integrating all the data and uploading it to a cloud server, beekeepers can monitor hive conditions remotely, even from a distant location.

2. System Design

As shown in Figure 1, this is the system architecture diagram of the present research.

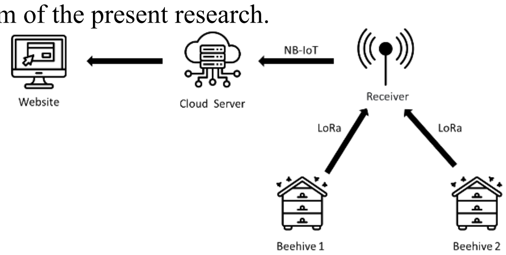

Fig. 1 The overall system architecture diagram.

As shown in Figure 2, this is the hardware circuit design diagram for the beehive. It utilizes temperature and humidity sensors, FSR, a laser module, Light Intensity Sensor, servo motor, and LoRa module to enable the operation of the beehive and data transmission.

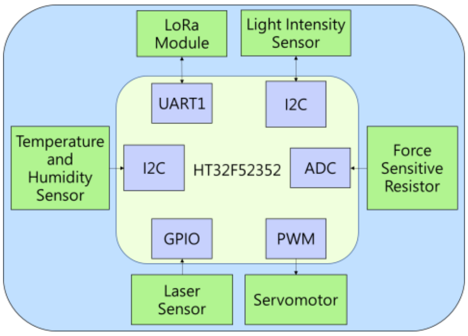

Fig. 2 The hardware schematic for the beehive.

As shown in Figure 3, this is the hardware circuit diagram for the receiver. It uses the LoRa module, NB-IoT module, GPS module, MPPT solar controller and temperature and humidity module to enable the receiver's positioning and data transmission capabilities.

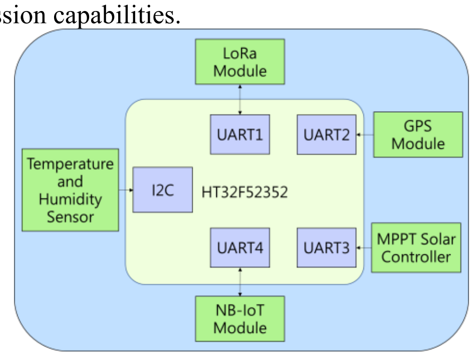

Fig. 3 The hardware schematic for the receiver.

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3 System Validation and Results

As shown in Figure 4, this is the physical diagram of the beehive used in this study.

a

b

Fig. 4 The physical diagram of the beehive
(a) The physical diagram of internal circuit of beehive.
(b) The actual breeding diagram of the intelligent Beehive IoT system.

In order to simulate the weight change of honey on the honeycomb sheet, the standard weight was used as the verification weight in this study. The measurement results are shown in Figure 5, where the X-axis represents the ADC value measured by the FSR, and the Y-axis represents the weight of the standard weight placed on the honeycomb sheet. The mathematical function of the parabola can be used to obtain the mathematical formula for converting the ADC value into the actual weight. When the weight is enough, the beekeeper will go to harvest, which can reduce the impact of spraying drugs to confirm the weight in the traditional breeding process and optimize the use of manpower.

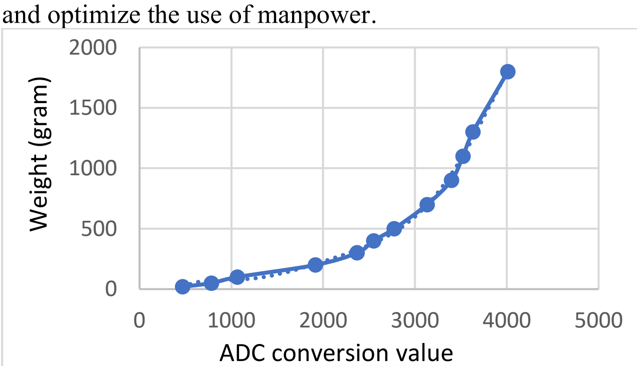

Fig. 5 Simulation of the honey weight variation curve.

In this study, data is uploaded to the cloud server via NB-IoT, and the collected data is visualized into report formats using a web page. This allows beekeepers to have a clear understanding of the Beehive's condition changes. As shown in Figure 6, the Cloud Server Web Interface.

3.1 Inside/Outside Temperature & Humidity (Celsius)

To accurately monitor the temperature variations inside the Beehive, temperature and humidity sensors were placed both inside and outside the Beehive for measurement. The measurement results are shown in the green box of Figure 6.

3.2 Rate of Change in Activity

To determine abnormal bee activity, Laser Sensor and Light Intensity Sensor were installed at the entrance of the Beehive to detect brightness. When the brightness falls below a certain value, it indicates bee movement. During the experiment, the experimenter simulated the bee's entry and exit by using their fingers to block the light source. The statistics of bee entries and exits from the Beehive are shown in the yellow box of Figure 6.

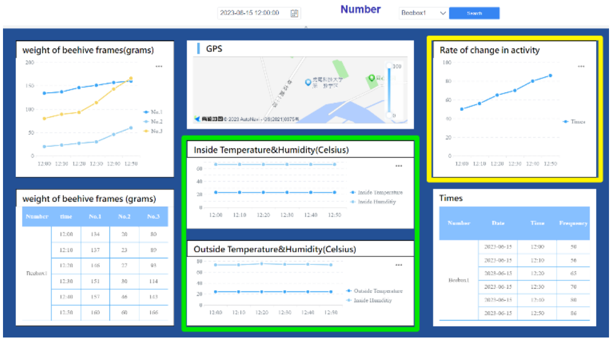

Fig. 6 Cloud Server Web Interface

4 Conclusion

This research is based on smart agriculture, with a focus on the application of smart sensing in beekeeping. Using IoT LoRa technology, a one-to-many beehive sensor network is established, which includes the collection of temperature and humidity data inside and outside the beehive, measurement of honey weight, and monitoring of bee activity. All sensor data is then uploaded to the cloud server using the NB-IoT module. With this setup, beekeepers can monitor honey production in real time and identify problems with honey production in a timely manner. In addition, to avoid forgetting the location of the hives, a GPS module is installed on the receiver, which sends the location data via NB-IoT to the cloud database for aggregation and analysis.

5 References

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